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Problem

- Vanilla policy gradients can take destructive large steps; TRPO controls them but requires second-order constrained optimization.
- Standard on-policy methods also discard each sample after a single gradient update.

2

Probability ratio

- $r_t(\theta) = \pi_\theta(a_t|s_t) / \pi_{old}(a_t|s_t)$
- The ratio measures how much the updated policy changes the probability of an action sampled by the behavior policy.

3

Clipped surrogate

- $L_{clip} = E[\min(r_t * A_t, clip(r_t, 1 - \epsilon, 1 + \epsilon) * A_t)]$
- Clipping pessimistically limits improvements when  $r_t$  leaves a neighborhood, yet keeps first-order optimization simple.

PPO

Proximal Policy Optimization

Schulman et al. | arXiv:1707.06347v2

Reuse on-policy rollouts for multiple minibatch epochs while clipping the policy ratio so updates stay useful without TRPO's constrained optimizer.

policy gradient

clipped objective

actor-critic

4

Training loop

- Actors collect trajectories; generalized advantage estimates feed several shuffled minibatch passes over the same data.
- A combined objective adds value-function error and an entropy bonus to the policy surrogate.

5

Evidence

MUJOCO + 49 ATARI GAMES

- PPO is tested on simulated locomotion and Atari and outperforms other online policy-gradient methods overall.
- It offers a favorable empirical balance of sample complexity, implementation simplicity and wall-clock time.

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Meaning & limits

THE PRACTICAL DEFAULT

- PPO made trust-region-like policy updates accessible and became a widely used baseline, including for language-model alignment.
- It remains on-policy and data hungry; clipping is not a formal trust-region guarantee and performance depends on tuning.